Quantitative Methods 1

Chapter 2: Measures of central tendency

Detailed solutions

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Exercise 4 — Centre and unusual observations

Seven customer baskets (euros): $16, 18, 18, 20, 22, 25, 91$.

1. Calculate the mode, median and mean.
2. Recalculate the mean excluding the last observation and compare.
3. Choose the most appropriate indicator of a typical basket.

Fully worked correction

Begin with the observed ordered table; do not calculate the mean before checking the observations and their frequencies.

x_i	16	18	20	22	25	91	Total
n_i	1	2	1	1	1	1	7
$n_i x_i$	16	36	20	22	25	91	210
Cum. count	1	3	4	5	6	7	–

1. Mode, median and arithmetic mean

$$\text{Mode} = 18 \quad (n_{18} = 2 > 1).$$

$$n = 7 = 2(3) + 1 \quad \Rightarrow \quad \text{Med} = x_{(4)} = 20.$$

$$\sum x_i = 16 + 18 + 18 + 20 + 22 + 25 + 91 = 210; \quad \bar{x} = 210/7 = 30.$$

2. Remove the last observation only for sensitivity analysis

$$S_{-91} = 210 - 91 = 119 = 16 + 18 + 18 + 20 + 22 + 25.$$

$$\bar{x}_{-91} = 119/6 = 19.8333, \qquad 30 - 19.8333 = 10.1667.$$

3. Explain indicator selection

The 91-euro basket shifts the mean upward while the median remains near the central observations. Median EUR 20 better represents an ordinary basket. The mean EUR 30 remains the relevant amount for reconciling total turnover ($7 \times 30 =$ EUR 210). Do not remove EUR 91 from actual revenue reporting.

Additional frequency check and decision-oriented explanation

Basket (EUR)	Count	Value × count	Ascending count	Ascending share
16	1	16	1	$1/7 = 14.29\%$
18	2	36	3	$3/7 = 42.86\%$
20	1	20	4	$4/7 = 57.14\%$
22	1	22	5	$5/7 = 71.43\%$
25	1	25	6	$6/7 = 85.71\%$
91	1	91	7	$7/7 = 100\%$
Total	7	210	–	100%

The fourth ordered basket is EUR 20, so EUR 20 is the median. The repeated observation EUR 18 is the mode. For describing an ordinary transaction, the median resists the unusually large EUR 91 basket; for accounting purposes, the arithmetic mean preserves the EUR 210 total. Omitting EUR 91 from the original turnover would be incorrect: its exclusion here is a sensitivity check only.

Exercise 5 — Selecting the appropriate average

A company has grades $12, 15, 17$ weighted $20\%, 30\%, 50\%$. Its revenue grows by $+8\%, -5\%, +10\%$ over three years. A courier travels 80 km at 40 km/h, then 80 km at 80 km/h.

1. Calculate the weighted average grade.
2. Calculate the compound average annual growth rate.
3. Calculate the average speed over the 160 km.
4. Explain why the ordinary arithmetic mean is not appropriate in every situation.

Fully worked correction

The starting calculation table identifies the aggregation rule for each situation.

Situation	Observations	Correct mean
Grades	$12; 15; 17;\ 20\%; 30\%; 50\%$	Weighted arithmetic
Growth	$+8\%; -5\%; +10\%$	Geometric
Speed	80 km at 40; 80 km at 80	Harmonic

1. Weighted grade: compute each weighted contribution

$$\sum w_i = 0.20 + 0.30 + 0.50 = 1; \quad \bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}.$$

$$0.20 \times 12 = 2.40; \quad 0.30 \times 15 = 4.50; \quad 0.50 \times 17 = 8.50.$$

$$\bar{x}_w = 2.40 + 4.50 + 8.50 = 15.40.$$

2. Geometric annual growth: compound wealth

$$F_1 = 1 + 0.08 = 1.08, \quad F_2 = 1 - 0.05 = 0.95, \quad F_3 = 1 + 0.10 = 1.10.$$

$$F = 1.08 \times 0.95 \times 1.10 = 1.026 \times 1.10 = 1.1286.$$

$$(1+g)^3 = 1.1286 \quad \Longrightarrow \quad g = 1.1286^{1/3} - 1 \approx 0.04115 = 4.12\%.$$

Verification: three consecutive years at the same constant annual rate multiply the starting revenue by 1.1286. In contrast, the simple mean of the three growth percentages equals (8-5+10)/3 = 4.33%, which ignores compounding.

3. Speed: time is distance divided by speed

$$t_1 = 80/40 = 2 \text{ h}, \quad t_2 = 80/80 = 1 \text{ h}.$$

$$D = 80 + 80 = 160 \text{ km}; \quad T = 2 + 1 = 3 \text{ h}; \quad \bar{v} = 160/3 = 53.3333 \text{ km/h}.$$

$$H = \frac{2}{1/40 + 1/80} = \frac{2}{3/80} = \frac{160}{3} = 53.3333.$$

The arithmetic mean of 40 and 80 equals 60 km/h, but would predict a travel time of $160/60 = 2.6667$ hours rather than the observed 3 hours.

Preparation tables and interpretation

Grade	Weight	Weighted contribution	Check
12	0.20	$12(0.20) = 2.40$	20%
15	0.30	$15(0.30) = 4.50$	30%
17	0.50	$17(0.50) = 8.50$	50%
Total	1.00	15.40	100%

The grade average is 15.40 on the original grade scale. The revenue growth factors must be multiplied: $1.08 \times 0.95 \times 1.10 = 1.1286$, and the equivalent constant annual growth is $1.1286^{1/3} - 1 \approx 4.12\%$. For the two travel legs, the times are $80/40 = 2$ hours and $80/80 = 1$ hour: $160/3 = 53.33$ km/h. A manager choosing a single "average" without asking what is being averaged would report incorrect growth and speed indicators.

Exercise 6 — Typical price and a pricing decision

Use the 150-consumer portable solar charger distribution from Exercise 3. The full table is reproduced below so that this worksheet can be used independently.

Class (euros)	12–18	18–24	24–30	30–36	36–45	45–60	60–75
Count n_i	6	14	28	42	30	20	10

1. Find the modal class using densities.
2. Find the median and quartiles by linear interpolation.
3. Estimate the mean using class midpoints.
4. Find the highest price affordable to at least 60% of consumers.

Fully worked correction

Initial statistical table. The width is the difference between boundaries, the midpoint is their mean, the density is the count divided by the width, and the cumulative count is the running sum. Quantiles and the mean below are estimates because the original prices within each interval are not observed.

Class	n_i	Width	Density	Midpoint	$n_i c_i$	N_i^+	N_i^-
$[12, 18[$	6	6	1.0000	15	90	6	150
$[18, 24[$	14	6	2.3333	21	294	20	144
$[24, 30[$	28	6	4.6667	27	756	48	130
$[30, 36[$	42	6	7.0000	33	1386	90	102
$[36, 45[$	30	9	3.3333	40.5	1215	120	60
$[45, 60[$	20	15	1.3333	52.5	1050	140	30
$[60, 75[$	10	15	0.6667	67.5	675	150	10
Total	150	–	–	–	5466	–	–

1. Modal class

The seven count densities are computed separately:

$$d_1 = 6/(18-12) = 1, \qquad d_2 = 14/(24-18) = 2.3333,$$
$$d_3 = 28/(30-24) = 4.6667, \qquad d_4 = 42/(36-30) = 7,$$
$$d_5 = 30/(45-36) = 3.3333, \qquad d_6 = 20/(60-45) = 1.3333,$$
$$d_7 = 10/(75-60) = 0.6667.$$

The highest density is 7 customers per euro, for $[30, 36[$. This is the modal class. When widths differ, raw counts are not comparable as histogram heights.

2. Linear interpolation: derivation and every substitution

Consider a target fraction p of all N observations. The target cumulative rank is $r = pN$. The class containing the target has lower bound L, upper bound U, width $a = U - L$, count n, and C observations strictly below the class. Under the *uniform-within-class* approximation, the within-class proportion of the interval to the left of the quantile q equals the within-class

proportion of observations required to reach r:

$$\frac{q-L}{U-L}=\frac{r-C}{n}.$$

Multiplying both sides by $U-L=a$ gives $q-L=a(r-C)/n$, hence

$$\boxed{q=L+a\frac{pN-C}{n}}.$$

We must identify the interval from the cumulative counts $6, 20, 48, 90, 120, 140, 150$ *before* substituting numbers.

First quartile, Q_1 (25%). $r=0.25(150)=37.5$. Since $20<37.5\leq 48$, the class is $[24,30[$. Thus $L=24$, $U=30$, $a=6$, $C=20$, $n=28$. The required within-class fraction is $(37.5-20)/28=17.5/28=0.625$:

$$Q_1=24+(30-24)\frac{37.5-20}{28}=24+6(0.625)=24+3.75=\boxed{27.75 \text{ EUR}}.$$

Median, Me (50%). $r=0.5(150)=75$. Since $48<75\leq 90$, the class is $[30,36[$. Therefore $L=30$, $U=36$, $a=6$, $C=48$, $n=42$, and $(75-48)/42=27/42=0.642857$:

$$Me=30+6\frac{75-48}{42}=30+\frac{162}{42}=30+3.857143=\boxed{33.8571 \text{ EUR}}.$$

Third quartile, Q_3 (75%). $r=0.75(150)=112.5$. Since $90<112.5\leq 120$, use $[36,45[$, with $L=36$, $a=9$, $C=90$, $n=30$. The fraction equals $22.5/30=0.75$:

$$Q_3=36+9\frac{112.5-90}{30}=36+9(0.75)=\boxed{42.75 \text{ EUR}}.$$

The central 50% has an estimated spread $IQR=Q_3-Q_1=42.75-27.75=15$ euros.

3. Approximate arithmetic mean: every class contribution

Midpoints are the averages of the two endpoints:

$$\begin{aligned}
c_1&=(12+18)/2=15, & n_1c_1&=6(15)=90,\\
c_2&=(18+24)/2=21, & n_2c_2&=14(21)=294,\\
c_3&=(24+30)/2=27, & n_3c_3&=28(27)=756,\\
c_4&=(30+36)/2=33, & n_4c_4&=42(33)=1386,\\
c_5&=(36+45)/2=40.5, & n_5c_5&=30(40.5)=1215,\\
c_6&=(45+60)/2=52.5, & n_6c_6&=20(52.5)=1050,\\
c_7&=(60+75)/2=67.5, & n_7c_7&=10(67.5)=675.
\end{aligned}$$

$$\sum n_ic_i=90+294+756+1386+1215+1050+675=5466, \qquad \bar{x}\approx\frac{5466}{150}=\boxed{36.44 \text{ EUR}}.$$

4. Maximum price affordable to at least 60%

Let W be a customer's maximum willingness to pay and p the listed price. The affordability condition is $P(W \geq p) \geq 0.60$, equivalent to $P(W < p) \leq 0.40$. The limiting price is therefore the *40th percentile*, not the 60th. Its rank is $r = 0.40(150) = 60$. Since $48 < 60 \leq 90$, it belongs to $[30, 36[$, with $L = 30$, $a = 6$, $C = 48$, $n = 42$:

$$P_{40} = 30 + 6\frac{60 - 48}{42} = 30 + \frac{72}{42} = 30 + 1.714286 = \boxed{31.7143 \text{ EUR}}.$$

At this estimated threshold, the cumulative proportion below it is 40% and the complementary proportion is 60%. Under the interpolation model, the maximum price is about EUR 31.71 (when prices are restricted to cents, rounding down preserves the target under the model). It is *not* guaranteed by the grouped data alone: the actual positions within the interval are unknown.

Commercial interpretation and limits of the approximation

The modal interval $[30, 36[$ contains the highest number of customers *per euro of interval width*. The estimated median EUR 33.86 means approximately half of stated maximum prices lie below it, assuming values are uniformly distributed within their reported class. $Q_1 \approx 27.75$ and $Q_3 \approx 42.75$ delimit an estimated middle half of the distribution. To ensure at least 60% can afford the product, use the lower 40th percentile: $P(W \geq p) \geq 0.60$ means $P(W < p) \leq 0.40$. This is a descriptive willingness-to-pay threshold, not a revenue-maximizing price; purchase probability, costs and demand responses are not identified.